

CoCo: An AI Voice Chat Enabled Information Kiosk For Columban College Inc.

A Design Project
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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

INTRODUCTION

In today's rapidly advancing digital generation, educational institutions are always in pursuit of innovative solutions of enhancing information access and communication. One of the new trends in the industry is the design of interactive information kiosks with the integration of artificial intelligence (AI) features. These are self-service information services in the model of interactive kiosks that provide quick and precise answers to queries from students, employees, and especially from freshmen and visitors. With the integration of voice chat through AI, the kiosk not only provides instantaneous responses but also a more natural, user-friendly experience, bridging the gap between human language and machines.

On a school campus, timely and reliable access to information is crucial to academic productivity, administrative operations, and overall satisfaction of students. Traditional information desks or written or printed notices such as bulletin boards have limitations like staff dependency or infrequent updates. An AI-powered information kiosk eliminates these limitations by providing an always accessible interface offering information on class schedules, school rooms and facilities, announcements, and other school services. Its voice interaction feature makes it more accessible, supporting even those who struggle with digital interfaces, thereby improving campus accessibility and inclusivity.

In an educational setting, students, most of the times visitors are having difficulty in looking for information real-time. The access of information is often

neglected due to outdated method of dissemination. Bulletins are not in action since the students do not usually see what is on display. Information is often distributed through bulletins which being neglected by most students. Students nowadays are not looking into bulletin boards since they prefer more advanced technology for information access, especially those interactable. Moreover, some of the students and visitors are too shy to ask inquiries to other people like campus guards, professors, and leaders. Sometimes the staff or faculty in the campus are too busy to entertain all student inquiries and asking the same question repeatedly can also become an exhausting and time-consuming burden.

To address these challenges, the design and deployment of an Information Kiosk with AI voice chat are aimed at modernizing the distribution of information in the school environment of Columban College, Inc. The project highlights the importance of integrating new and innovative technologies to push the institution towards efficiency, innovation, and digitalization, supporting its vision of being an innovative institution of learning. Moreover, it aims to demonstrate the capability of AI-based systems in enhancing the learning experience by developing the smart campus environment where information is made readily available, interaction is simplified, and the communication barrier is minimized.

The development of an Information Kiosk with AI voice chat aims to provide students staffs, and visitors with quick and easy access to essential information within the Columban College Inc. The kiosk will offer details directories such as faculty locations, canteen and library locations, building layouts, schedules, persons in charge, available facilities, school rules, laboratory rules, events, information included in the student manuals, student uniform guidelines, dos and don'ts, vision, mission and core

values of the school, school hymn, and so on. By centralizing this information, the kiosk addresses the limitations of traditional bulletin boards and staff inquiries, providing real-time, accessible guidance tailored specifically to the Engineering Department.

STATEMENT OF THE PROBLEM

Due to the challenges encountered by the students and visitors of Columban College Inc. in accessing timely information about the school, CoCo: An Information Kiosk using AI Voice Chat was proposed.

The study determined the different difficulties experienced by students and visitors in the existing manual method of asking questions or relying on bulletin boards and limited staff assistance.

Specifically, the following problems is sought to be understood:

1. How may CoCo be designed and developed in terms of hardware components to support voice input and output function?
2. How may the system be implemented to accurately process voice input into reliable events to ensure smooth interaction between the user and ai system?
3. How may the system handle repetitive inquiries to improve efficiency for answering inquiries?
4. How may the proposed information kiosk be integrated with AI voice chat to provide accurate and relevant responses to students and visitors?
5. How can the system be tested and implemented effectively within the school environment?

OBJECTIVES OF THE STUDY

The main goal of this study is to design and develop an Information Kiosk using AI Voice Chat for Columban College Inc. that can provide accurate, real-time, and accessible campus-related information to students, visitors, and staff.

Specifically, this study aims to:

1. Design and develop the hardware components of CoCo to support both voice input and output functions.
2. Design and develop an information kiosk that is capable of processing voice queries and replies with speech accurately and real-time.
3. Design and develop software capable of delivering intelligent responses based on institutional facts.
4. Integrate the AI voice chat feature into the information kiosk to provide accurate and relevant answers to users.
5. Test and evaluate the system's performance in an actual setting.

SIGNIFICANCE OF THE STUDY

- A. Students.** In this study, the result will help them by seeing how AI technology can solve real-world problems. Also, may inspire students to innovate and contribute to technology research.
- B. Instructors/Professors.** This study is significant to engineering instructors and professors in terms of modern technology and emerging technology in today's technology-based world. This will help them introduce platform in the prospect of emerging technology and AI subjects.
- C. Community.** This study will be beneficial for the community in terms of creating an information kiosk with ai voice chat integration, which will help people who want to make better systems.
- D. Institution.** This study will help the institution to see how AI-powered information kiosk can be used in real life. It can also serve as a reference for future research and school projects related to AI voice chat integration.
- E. Future Researchers.** This study will serve as a foundation for future researchers who want to improve or expand the use of AI-integrated kiosks. It will also guide them in developing better systems that can serve both academic and non-academic needs.

SCOPE AND LIMITATIONS

The study focuses on the development of an AI-powered Information Kiosk using voice chat that will serve as a modernized information kiosk for Columban College, Inc. The system aims to address the limitations of traditional information desks and bulletin boards by providing an interactive, self-service platform that delivers quick and accurate responses to students, faculty, and visitor queries.

The following are the scopes of the proposed study:

1. The system provides an AI voice chat-bot capable of answering frequently asked questions such as class schedules, school rooms and facilities, and general school services.
2. The system is equipped with 24-inch Monitor and serves as visual display where responses are shown in real-time.
3. The system utilizes OpenAI-powered chat with speech-to-text and text-to-speech features to enable natural voice-driven interaction between the kiosk and the user.
4. The system contains curated information specific to the Engineering Department, including organizational charts, faculty lists and notable achievements.
5. The system operates based on the data provided by the institution, ensuring that responses are aligned with school policies and verified information.
6. The kiosk is designed for voice and visual interaction, offering both audio output and real-time display to enhance user experience.

There is no system that is ideal; thus, this project has its limitations which include:

1. The system requires stable internet connectivity for AI voice interaction; thus, reliability and prompt replies relies strongly on Campus Wi-Fi. Any instability may produce erroneous.
2. The system is capable of English queries. Any Filipino, Taglish, or other languages are not supported.
3. The school campus related knowledge of the AI Bot's is limited to the information provided by institutions to the system's database. Any queries outside the given data cannot be answered.
4. The system's testing and validation are limited to the Engineering Department; therefore, some information is more complete for this department while other departments may have partial or incomplete datasets.
5. The system cannot automatically update or verify new institutional information. Any changes in schedules, announcements, or faculty details must be manually updated by the administrators.
6. Background noise or crowded environments can affect the accuracy of speech recognition and system performance.
7. The kiosk cannot provide personalized services such as student records, grades, or account-specific transactions.

DEFINITION OF TERMS

Information Kiosk – A small digital station or booth where people can get information about school services, events, and announcements.

Artificial Intelligence (AI) – A computer system that can think, learn, and respond like humans. It helps the system understand questions and give answers.

Voice Recognition – A technology that allows the computer to understand spoken words and turn them into commands or responses.

Chat Integration – A feature that combines AI with a chatting system, allowing students to ask questions and receive instant answers.

User-Friendly – A design that is simple, easy to use, and accessible for everyone, even without technical knowledge.

Open-AI – A company that develops advanced artificial intelligence technologies, including tools that help computers understand and generate human-like text.

AI Chatbot – A computer program that uses artificial intelligence to talk with users, answer questions, and provide information like a real person.

CONCEPTUAL FRAMEWORK

The researchers conceptualized a workflow design using the input-process-output framework to be followed upon the conduct of the research as shown in the Figure 1. The framework includes the mechanisms for processing and analyzing voice queries and delivering intelligent responses. This involves voice recognition, AI processing, and multimedia output systems that convert user inquiries into accurate and accessible information.

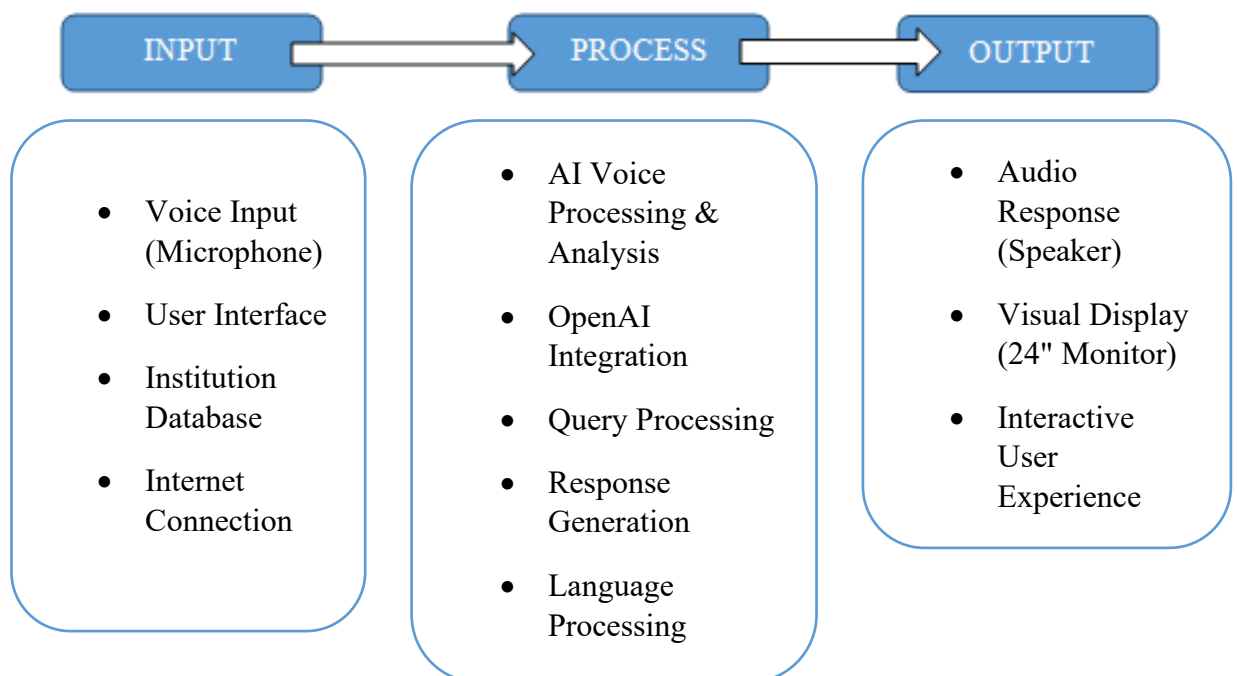


Figure 1. *Conceptual Framework*

Figure 1 shows a summary of the functions of the CoCo system. The input indicates where the data is obtained from, such as voice input through microphone, user interface, institutional database, and internet connection. The process involves AI voice processing and analysis, ChatGPT integration, query processing, response generation, and language processing. Finally, the output shows audio response through speaker, visual display on 24-inch Monitor, and interactive user experience.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

RELEVANCE OF THE RELATED LITERATURE

In schools and universities, access to accurate and timely information is a constant challenge that requires innovative solutions to improve efficiency and user convenience. Traditional methods, such as bulletin boards, printed announcements, or manual inquiries with staff, often result in delays, miscommunication, and student inconvenience, especially during busy academic periods. As a response, various studies have explored the use of information kiosks and artificial intelligence (AI) to provide more accessible, reliable, and user-friendly information services. This section reviews related literature, studies, and theories that support the development of CoCo: An Information Kiosk with AI Voice Chat Integration at Columban College Inc. Engineering Department.

The University of Southeastern Philippines (USEP) deployed an AI chatbot named SePhi to improve student and stakeholder engagement. The chatbot utilizes Natural Language Processing (NLP) and artificial intelligence to provide accurate and real-time responses to inquiries. It was integrated with Facebook Messenger and is planned for expansion to the official university website. The project showed that incorporating AI in education can improve accessibility to information and streamline communication between the institution and its students (**University of Southeastern Philippines, 2023**).

In Pangasinan, the development of an information kiosk system was undertaken to promote tourism by providing comprehensive information about

accommodations, dining, attractions, events, and culture in multiple languages. The system was designed to be administered by the local IT office and strategically located in accessible tourist areas without requiring online transactions for optimal usability. The research emphasized the importance of multilingual support and offline functionality to ensure broad accessibility regardless of internet connectivity. The kiosk system was developed to benefit the province of Pangasinan, Department of Tourism, tourists, and developers by providing centralized access to essential tourism information.

In Northern Philippines, Mariano Marcos State University (MMSU) collaborated with the Department of Science and Technology (DoST) to establish digitized information kiosks in the main library, providing internet-free access to thousands of science and technology resources. The initiative aimed to democratize access to academic materials and research resources without requiring internet connectivity, ensuring that students and researchers could access critical information regardless of network availability. The system was specifically designed to support academic research and learning by providing quick access to digitized science and technology databases, journals, and educational materials. The strategic placement in the university library maximized accessibility for students and faculty members.

According to **Raj (2024)**, information kiosks are revolutionizing customer experience by offering instantaneous, self-service access to product details, wayfinding, and personalized content. The article emphasizes how kiosks reduce reliance on staff, cut down wait times, and provide customers with more autonomy in obtaining needed information. Key features include real-time updates, AI-powered personalization, interactive wayfinding, and dynamic content delivery that adapts based on user

behavior. This literature is relevant to CoCo's development because many of the same benefits apply in educational settings: students, visitors, and faculty often need fast, accurate information and an AI voice-enabled kiosk can replicate many of the advantages seen in retail kiosks. By integrating real-time voice interactions, CoCo can further enhance accessibility, reduce friction for users, and improve user satisfaction in Collumban College's Engineering Department, mirroring the transformation anticipated in the retail space.

RELEVANCE OF THE RELATED STUDIES

When AI chat assistants were first introduced, they aimed to provide faster and more accessible ways of addressing user concerns. Over the years, many institutions have adapted these technologies to improve information services, especially in academic settings.

In the Philippines, **Corpuz (2025)** developed the Student Academic Record Evaluation Kiosk (SAREK) for the College of Engineering, Architecture, and Technology at Isabela State University. The kiosk was designed to allow students to view and evaluate their academic records without relying heavily on administrative staff. Through the system, service delivery was improved, waiting times were reduced, and students were able to access information more efficiently. The evaluation results highlighted the kiosk's effectiveness in providing student services and emphasized its role in reducing workload for faculty and staff. Furthermore, the study discussed the importance of usability testing and user feedback in ensuring system effectiveness, which can serve as a guide for future kiosk-based solutions.

In Cavite, **Ponce et al. (2018)**, the researchers conducted a study on the development of an intelligent kiosk system designed to assist users in retrieving information more efficiently. The research highlighted that traditional information dissemination methods, such as printed materials and manual inquiries, often led to delays and inefficiencies in academic institutions. To address these challenges, the proposed kiosk utilized multimedia features and a user-friendly interface to provide fast, accurate, and accessible information to students and visitors. The system's design emphasized convenience and adaptability, allowing users to access needed services without relying on administrative staff. This study is relevant to CoCo's development because it shows how information kiosks can streamline service delivery, improve user satisfaction, and serve as a foundation for integrating advanced features such as AI-driven voice chat to further enhance accessibility and user interaction within Columban College Inc. Engineering Department.

Blanco et al. (2022) developed an information kiosk system for ABE Manila to address the lack of directional signs and time-consuming process of obtaining accurate information about campus locations. The researchers identified that students and visitors frequently struggled to navigate the campus efficiently, leading to frustration and delays in accessing needed information. The proposed interactive information kiosk was designed to provide independent access to campus data, enabling users to find locations without requiring assistance from others. The system aimed to make information access more inclusive by supporting users with varying literacy levels through intuitive touchscreen interfaces. This study directly relates to CoCo's development as it demonstrates the practical need for self-service information systems and validates the effectiveness of interactive kiosks in addressing navigation and

information accessibility challenges that are similarly present at Columban College Inc. Engineering Department.

In Metro Manila, **Sabado et al. (2022)** presented a comprehensive kiosk system for Micro Asia College of Science and Technology Inc. The research focused on developing a user-friendly interface that could accommodate the diverse needs of students, faculty, and visitors within an academic environment. The system incorporated touchscreen technology to provide interactive access to institutional information, class schedules, faculty directories, and campus services. The researchers emphasized the importance of creating an intuitive design that would reduce the burden on administrative staff while improving the overall user experience.

Chen et al. (2023) explored the design and implementation of AI-powered student assistant chatbots to support classroom learning and improve student success. Their study emphasized that AI chatbots, when carefully designed, can provide timely academic support, encourage engagement, and improve overall learning experiences. By integrating natural language processing and conversational AI, the chatbots were able to assist students with inquiries ranging from course content to administrative tasks, thereby reducing reliance on faculty and staff for routine information. The research highlighted that effective chatbot design must consider personalization, accessibility, and seamless interaction to meet diverse student needs.

Udvarhelyi (2023) developed self-service kiosk design guidelines for a digital retail ecosystem, focusing on how kiosks can improve customer engagement and accessibility within omnichannel environments. The study identified critical design considerations such as encouraging usage, providing flexibility for diverse users, reducing cognitive load, and ensuring accessibility and inclusivity. Using qualitative

methods, including usability testing and thematic analysis, the research showed that kiosks are most effective when they personalize user experience, integrate with broader digital systems, and are strategically placed for maximum visibility and use. This study is relevant to CoCo's development as it demonstrates the importance of designing kiosks with user-centered principles, which can guide the Engineering Department of Collumban College in creating a kiosk that is both functional and inclusive. The emphasis on personalization and accessibility directly supports CoCo's innovation of integrating AI voice chat to enhance interaction and usability for students and staff.

In Canada, **Falahati et al. (2023)** introduced HERMES, an AI-powered public health kiosk designed to provide personalized over-the-counter medication recommendations and health education in public spaces. The kiosk used advanced AI models, including Graph Attention Networks and federated learning, to ensure accurate, context-aware recommendations while preserving user privacy. Key features included multilingual support, accessibility functions such as voice commands and Braille compatibility, and iterative usability testing to ensure inclusivity across diverse demographics. The study concluded that AI-driven kiosks can expand access to critical services, empower users, and improve efficiency in high-demand settings. Just as HERMES provides advanced healthcare services through AI, CoCo aims to provide academic and institutional support, showing how kiosks with intelligent systems can be adapted to education to improve student engagement and service delivery.

In Turkey, **Ekşioğlu et al. (2018)** developed an interactive information kiosk with a user-centered approach (UXD principles) that efficiently supports students in a university campus, improving access to information, reducing reliance on help desks, and enhancing overall campus experience and satisfaction. The authors of this study

concluded that an information kiosk is needed at Boğaziçi University. This study highlights how UXD principles are important in designing user-friendly and efficient campus information systems that meet the actual needs of students and visitors which serves as a foundation and guiding framework in the development of CoCo. Moreover, this study will serve as one of the references in UI design of CoCo and selection of data to be included in the kiosk.

Shanmugasundaram (2025) conducted study about a comprehensive review and analysis of AI-driven chatbot kiosks, focusing on their technical architecture, natural language processing mechanisms, industry-specific applications, user experience, and future trends, in order to inform AI industry leaders and technology architects about the opportunities and challenges of implementing intelligent kiosk systems in real-world settings. This study that gives comprehensive description and recommendation about AI-driven chatbot kiosks will serve as a key theoretical and practical reference and guide in the development of CoCo.

A study about developing a personal AI voice assistant demonstrates the potential of AI to enhance human-computer interaction and daily life (**Mhetre et al., 2024**). This research highlights how advanced natural language processing, task automation, and local voice command processing can create efficient, convenient, and privacy-conscious user experiences. In connection to CoCo, the study contributes methodologies and best practices for integrating AI-driven voice interaction into a physical kiosk, enabling students, visitors, and staff to access campus information, navigate facilities, and interact with the system naturally and efficiently. By adopting similar approaches, CoCo can offer a more responsive, user-friendly, and inclusive interface while addressing accessibility concerns.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter presents the research methodology employed in the design and development of CoCo: An AI Voice Chat Enabled Information Kiosk For Columban College Inc. It outlines the research design, population design and sampling technique, data collection procedures, research instruments, data analysis methods, and the system methodology.

Research Design

The study uses an experimental research design, where the researcher manipulates one or more independent variables and measures their effect on dependent variables to establish cause and effect relationships. The Office of Research Integrity (2005) defines experimental research as “a study in which a treatment, procedure, or program is intentionally introduced, and a result or outcome is observed.” In this study, the CoCo system was tested by applying controlled changes and observing their outcomes to assess its overall performance and user response. This study aims to design, develop, and implement a working prototype of an information kiosk using Raspberry Pi 5, OpenAI module and IoT sensors and actuators. The system will allow users to access information and interact with the kiosk using voice commands, even when staff assistance is unavailable, through connection to the internet to enable real-time updates and AI-driven responses, ensuring that users receive accurate and timely information efficiently.

Population Design and Sampling Technique

The purposive sampling technique will be used for this study. In purposive sampling, participants are selected because they meet specific criteria relevant to the research objectives. The target respondents of this study will include students, and staff such as guards from Columban College Inc only. Also, engineering students will interact with the kiosk prototype during the testing phase. Their feedback will serve as the basis for evaluating the system's usability, efficiency, and reliability in providing accurate information.

Statistical Tool

Descriptive statistics will be utilized to analyze the data collected from the respondents. Frequency counts, percentages, and weighted means will be applied to the survey responses, particularly in the Likert-scale items, to evaluate the performance of the system in terms of accuracy, ease of use, responsiveness, and overall user satisfaction. The use of descriptive statistics allows the researchers to summarize the responses in a clear and measurable manner, providing insights into the system's effectiveness.

Data Collection Procedure

The data collection procedure will follow several stages. First, a Google Form survey will be conducted to identify the most frequently asked questions (FAQs) to students of Columban College Inc., including inquiries about class schedules, facilities, directories, history, and student guidelines. Information from the school's official student manual will also be gathered to ensure that the AI chatbot provides accurate and institutionally approved responses. Since directories and certain rules are not

included in the student manual, the researchers will also gather information from across the campus regarding directories and rules/policies, such as regulations in laboratories, libraries, and other facilities. Additionally, one-on-one interviews will be conducted with campus security guards to collect information on the most common questions asked by visitors, ensuring that the AI system can address inquiries from both students and visitors. This comprehensive data will be used to build the initial knowledge base of the AI chatbot.

After gathering the FAQs and other campus information, the researchers will proceed with the design and development of the kiosk prototype, integrating the AI voice chat system with hardware components such as the display monitor and microphone. Once the prototype is completed, it will be tested in the Engineering Department. Students and staff will be invited to interact with the system by asking questions. Afterward, survey feedback questionnaires will be distributed to gather feedback regarding the system's usability, efficiency, and accuracy.

Research Instrument

The instruments that proponents used in collecting the data are observation checklist, prototype testing, and survey feedback questionnaires. An observation checklist will be developed to monitor the performance of the kiosk during its trial phase. This checklist will evaluate criteria such as the accuracy of responses, processing time of voice input, system reliability, and user-friendliness of the kiosk interface. The prototype testing will involve the actual use of the kiosk in a controlled environment where respondents can ask questions and observe how the system responds. Finally, survey questionnaires will be distributed to the respondents, which will include Likert-scale items to quantitatively measure usability, accuracy, and satisfaction.

Observation Checklist

An observation checklist will be developed to monitor the performance of the kiosk during its trial phase. This checklist will evaluate criteria such as the accuracy of responses, processing time of voice input, system reliability, and user-friendliness of the kiosk interface.

Evaluation Criteria

Criteria	Performance Indicators	Rating Scale
Response Accuracy	Correctness of information provided	1-5 (Poor to Excellent)
Voice Recognition	Ability to understand user queries	1-5 (Poor to Excellent)
Processing Time	Speed of response generation	1-5 (Very Slow to Very Fast)
System Reliability	Frequency of errors/crashes	1-5 (Frequent Issues to No Issues)
Interface Usability	Ease of navigation and interaction	1-5 (Very Difficult to Very Easy)

Table 1 *Evaluation Criteria*

During the observation, several technical issues were encountered that affected system performance. User behavior patterns were also noted, including how individuals interacted with the system and responded to errors or delays. Environmental factors, such as noise, and network stability, were observed to influence the system's functionality. Additionally, the frequency of successful versus failed interactions was recorded to assess the reliability and efficiency of the system.

Testing Procedure

To test the performance of this project, the proponents will conduct a testing that will determine the accuracy of responses, processing time of voice input, system reliability, and user-friendliness of the kiosk interface. The prototype testing will involve the actual use of the kiosk in a controlled environment where respondents can ask questions and observe how the system responds.

Step-by-Step Procedure:

Step 1: The participants will be briefed on the functionality of the kiosk.

Step 2: The respondents will activate the kiosk by saying “CoCo” and then perform specific queries.

Step 3: The proponents will monitor the user-system interaction throughout the testing session.

Step 4: The proponents will gather immediate feedback by giving survey feedback questionnaires containing Likert-scale items to the respondents after completing their queries.

Step 5: End of testing.

Survey Feedback Questionnaires

For additional data for system evaluation, survey feedback questionnaires will be administered after the participants have tested the system. These questionnaires aim to gather feedback regarding the system's usability, efficiency, and accuracy.

The survey feedback questions were rated using 5-point Likert scales. Responses to the survey were interpreted using the following scale:

Scale	Verbal Interpretation
4.20 – 5.00	Strongly Agree
3.40 – 4.19	Agree
2.60 – 3.39	Moderately Agree
1.80 – 2.59	Disagree
1.00 – 1.79	Strongly Disagree

Table 2 5- point Likert scales

The table shows the survey feedback questionnaire form designed for CoCo: An Information Kiosk with AI Voice Chat Integration at Columban College Inc. The figure illustrates the layout and structure of the printed questionnaire distributed to the respondents during the system testing phase. It includes sections for usability assessment, satisfaction assessment, and open-ended questions. Respondents were instructed to rate their experiences using a 5-point Likert scale, ranging from “Strongly Disagree” to “Strongly Agree” and from “Very Dissatisfied” to “Very Satisfied.” The form also provided space for optional respondent details, such as name and course, and invited participants to share qualitative feedback regarding their interaction with the CoCo system.

**Survey Feedback Questionnaires for CoCo: AN INFORMATION KIOSK WITH AI VOICE CHAT
INTEGRATION AT COLUMBAN COLLEGE INC.**

The purpose of this survey is to gather feedback about CoCo, an information kiosk equipped with AI voice chat integration. Your responses will help assess its usability, efficiency, and overall user experience. The information you provide will remain confidential and will be used solely for research and development purposes. Thank you for your valuable participation!

Name (optional): _____ Year and Course (optional): _____

Usability Assessment

Instructions: Rate your agreement with the following statements using the scale below:

- 5 = Strongly Agree
- 4 = Agree
- 3 = Neutral
- 2 = Disagree
- 1 = Strongly Disagree

	Statement	Rating
1	The kiosk was easy to use	1 2 3 4 5
2	The voice recognition worked accurately	1 2 3 4 5
3	The kiosk responded quickly to my queries	1 2 3 4 5
4	The information provided was accurate	1 2 3 4 5
5	The interface was user-friendly	1 2 3 4 5
6	I felt comfortable using the voice feature	1 2 3 4 5
7	The kiosk understood my questions clearly	1 2 3 4 5
8	I would use this kiosk again	1 2 3 4 5
9	The kiosk is better than traditional information desks	1 2 3 4 5
10	I would recommend this kiosk to others	1 2 3 4 5

Satisfaction Assessment

Instructions: Please rate your satisfaction with each aspect of the kiosk

	Aspect	Very Dissatisfied	Dissatisfied	Neutral	Satisfied	Very Satisfied
1	Overall experience	1	2	3	4	5
2	Response accuracy	1	2	3	4	5
3	System speed	1	2	3	4	5
4	Ease of use	1	2	3	4	5
5	Voice recognition	1	2	3	4	5

Open-ended Questions

1. What did you like most about the kiosk?
2. What aspects need improvement?
3. Did you encounter any difficulties? Please describe.
4. Additional comments or suggestions:

Figure 2. Survey Feedback Questionnaire

Statistical Treatment of Data

The data gathered from the survey feedback questionnaires and observation checklists will be analyzed using descriptive statistics to evaluate the performance and effectiveness of the CoCo information kiosk system. The following formula was used in the study:

Percentage:

$$\text{Percentage (\%)} = (f / N) \times 100$$

Where:

f = frequency of responses for a specific rating

N = total number of respondents

Weighted Mean:

$$\text{Weighted Mean (WM)} = \Sigma(f \times w) / N$$

Where:

- f = frequency of responses for each rating
- w = weight assigned to each rating (1 to 5)
- N = total number of respondents
- Σ = summation of all products

The interpretation of the weighted mean will be based on the following scale:

Scale Range	Verbal Interpretation
4.20 – 5.00	Strongly Agree
3.40 – 4.19	Agree
2.60 – 3.39	Moderately Agree
1.80 – 2.59	Disagree
1.00 – 1.79	Strongly Disagree

Table 3 5- point Likert scales

Standard Deviation:

$$SD = \sqrt{[\Sigma(x - \bar{x})^2 / N]}$$

Where:

SD = standard deviation

x = individual rating

$\bar{x}$ = mean rating

N = total number of respondents

Σ = summation

A lower standard deviation indicates that responses are closely clustered around the mean, suggesting consistency in user experience. A higher standard deviation suggests more varied opinions among respondents.

Success Rate Calculation:

For the observation checklist data, the success rate of interactions will be calculated to assess system reliability. This will measure the percentage of successful queries versus failed or unsuccessful attempts. The formula is:

$$\text{Success Rate (\%)} = (\text{Number of Successful Interactions} / \text{Total Number of Interactions}) \times 100$$

This metric will provide quantitative evidence of the system's operational reliability and effectiveness in responding to user queries.

Response Time Analysis:

The average processing time for voice input and response generation will be calculated using the arithmetic mean of all recorded response times during the testing phase. This will be expressed in seconds and will help evaluate the system's efficiency in terms of speed and responsiveness.

$$\text{Average Response Time} = \Sigma(\text{Response Time}) / \text{Number of Interactions}$$

The analyzed data will be presented using tables and figures to facilitate clear interpretation and understanding of the results. Frequency distribution tables will show the breakdown of responses for each criterion. Summary tables will display the computed weighted means, standard deviations, and verbal interpretations for each evaluation aspect. Charts or graphs may be used to visualize trends and patterns in the data, making comparisons across different criteria more accessible.

Ethical Consideration of CoCo

The development and deployment of CoCo, an AI voice chat-enabled information kiosk for Columban College Inc., introduces several ethical considerations that must be carefully addressed to ensure responsible use of the technology. One primary concern relates to the collection and processing of voice data. Since CoCo relies on audio input to interpret user queries, the system may unintentionally capture sensitive information, personal identifiers, or emotional cues embedded in a user's voice. Voice data, similar to biometric information, is highly personal and carries privacy risks if it is improperly stored or accessed. According to **CloudTalk (2024)**, voice-AI systems must implement strict data-protection mechanisms due to the possibility of unauthorized access or misuse of audio recordings. For CoCo, safeguarding user privacy will be a priority, which involves secure data handling practices, encryption, limited retention periods, and transparency regarding how voice data is managed throughout the system.

Another central ethical issue involves informed consent and transparency in the use of the kiosk. Users must be clearly informed that they are interacting with an artificial intelligence system rather than a human representative. According to **UNESCO (2021)**, AI systems should provide explicit disclosure of their nature, capabilities, and limitations to avoid misleading users or creating false impressions of human-like understanding. Since the kiosk may be used by students, staff, and campus visitors with varying levels of technological familiarity, it is essential to communicate this information in accessible, easy-to-understand language. Informed consent also requires informing users about how their voice inputs are processed, what data may be temporarily stored, and what protective measures are in place to secure this information.

Ensuring that users choose to engage with the kiosk voluntarily, without pressure or obligation, promotes their autonomy and strengthens trust in the system.

In addition, the issue of fairness and accessibility arises when implementing a voice-activated system in a diverse school environment. AI-based speech recognition technologies often perform unevenly across different accents, speech patterns, or linguistic backgrounds. **CloudTalk (2024)** notes that this may lead to unequal user experiences or unintended exclusion of individuals. For a college campus with diverse speech profiles, it is essential to ensure that CoCo functions reliably across a wide range of voices. Accessibility considerations must also be included, such as offering on-screen text input options for users who have speech impairments or prefer non-verbal interaction. These considerations help reduce bias and promote equal access to campus information and services.

Moreover, responsible implementation requires proper human oversight and accountability. Although CoCo is designed to automate information delivery, it must not fully replace human support, especially when addressing academic concerns, student services, or situations that require empathy and judgment. **AIHub (2021)** highlights the importance of maintaining human supervision in AI-mediated interactions to prevent misinformation, misinterpretation, or errors that may negatively affect users. Clear protocols must be established for correcting system mistakes, updating inaccurate responses, and ensuring that campus personnel can intervene when necessary. This helps maintain a transparent and accountable system that aligns with institutional values.

Finally, broader societal and institutional implications must be considered. The introduction of AI-based information services may unintentionally create unequal

access if certain groups within the college community benefit more from the technology than others. **Jobin, A. et al (2019)** highlight that AI systems can influence fairness and accessibility, particularly when users have differing levels of digital literacy or comfort with technology. Similarly, **Floridi and Cowls (2019)** emphasize the importance of designing AI tools that support social good while avoiding the reinforcement of existing inequalities. For CoCo, this means ensuring that the kiosk complements existing human support systems rather than replacing them, and that individuals who prefer traditional assistance or who have limited access to technology are not disadvantaged. By addressing these ethical concerns, the project promotes responsible AI development and equitable access to information for the entire campus community.

System Methodology – Iterative Model

The system will be developed using iterative software development life cycle approach. The proponents choose this approach mainly for its design that grants flexibility in requirements, allowing changes in requirements that can be accommodated in later iterations without affecting the entire system. In this way, the main features of the system can be implemented earlier, while other supporting features can wait for later iterations. Another advantage of using this approach is that it is beneficial for early detection of issues in which problems or missing requirements can be identified in the development process early since the system is built in evolving versions, allowing issues and changes to be addressed without waiting for the full system to be built. Furthermore, the iterative design allows better management of resources which allows efficient allocation of time and resources.

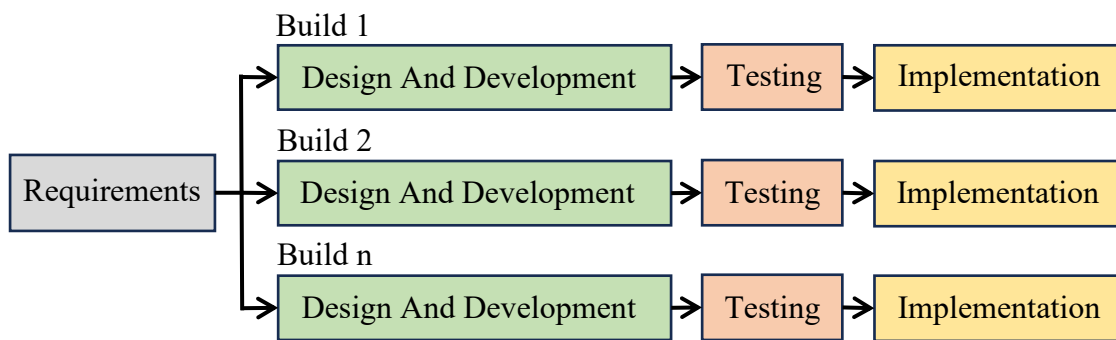


Figure 3 *Iterative Model*

Figure 3 shows iterative SDLC model which follows a cyclic approach where each phase is repeated in multiple iterations. The process starts with requirement analysis, followed by design, implementation, and testing. Each cycle produces a working version of the system that is refined based on feedback from the previous iteration. Testing results guide adjustments in the next cycle, improving functionality

and quality over time. The model emphasizes early detection of errors and continuous enhancement, allowing the system to evolve gradually while maintaining alignment with user requirements.

1. **Requirements:** The process in the iterative model starts with gathering system requirements for its goals, including the identification of the needed hardware and software components, necessary features, and user needs. The requirements are analyzed, prioritizing the main features in the earlier cycles and adding supporting features in the later iterations. Each cycle refines the requirements to match changing conditions. This ensures the system grows in alignment with meeting user needs.
2. **Design and Development:** After requirements are gathered, the design and development phase begins. The requirements are translated into models and system architecture. Each cycle develops features based on the current design. Coding produces a working version of the system. The process repeats with improvements or new features added in every cycle.
3. **Testing:** Each system build undergoes systematic testing, including functional checks, response time verification, and reliability assessment. The goal is to identify errors, defects, or areas needing improvement. Feedback from testing guides the adjustments in the next iteration, ensuring continuous enhancement of system quality. This cyclical testing ensures that each version of the kiosk becomes progressively more stable, reliable, and user-friendly.
4. **Implementation:** The tested version is deployed for use. The system runs in its current state and is observed in practice. Feedback from users and performance results guide the next cycle. Each implementation delivers a more complete system.

3.1 System Requirements

This section identifies the necessary requirements needed to develop and operate the CoCo. System requirements outline the essential specifications for both hardware and software to ensure that the kiosk functions efficiently and reliably. These requirements include the tools, platforms, devices, and technologies needed to support voice processing, AI response generation, data retrieval, and user interaction. By defining these requirements early in the methodology, the proponents ensure that the system is built on a stable, compatible, and well-supported foundation that meets the needs of its intended users.

Gantt Chart

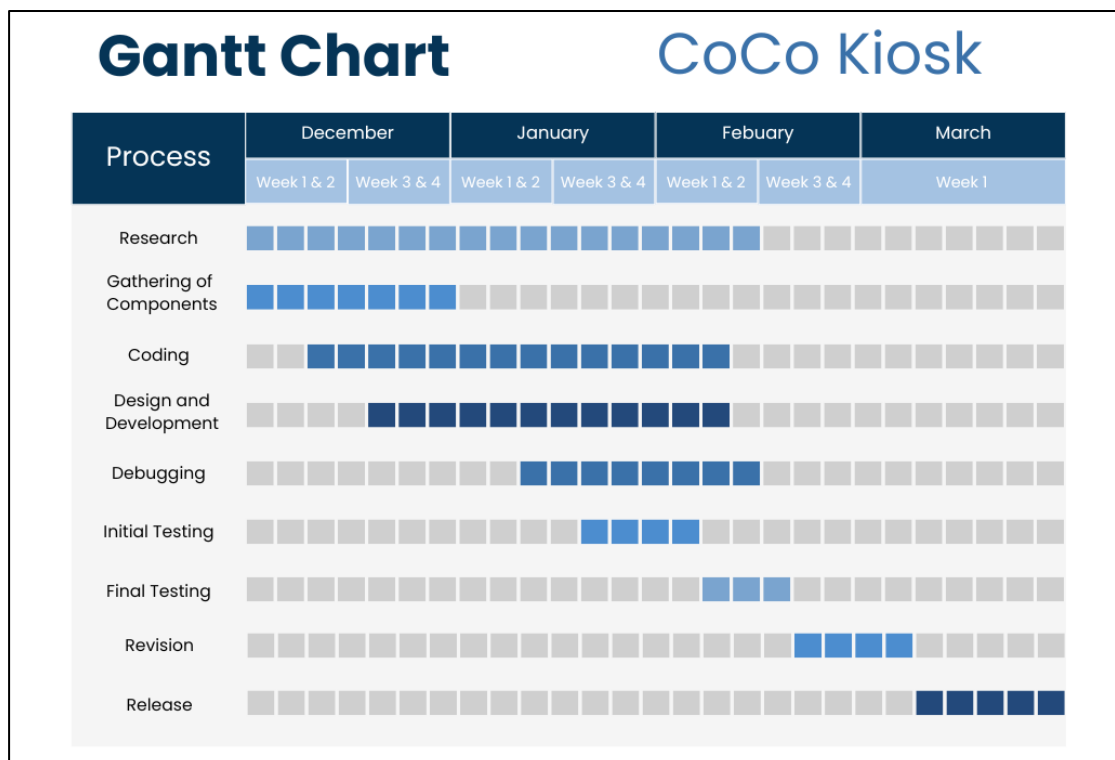


Figure 4 *Gantt Chart*

This Gantt Chart illustrates the project's procedural flow using an iterative method. It outlines the timeline of tasks leading up to the project's completion in March. The month of December focuses on research on deep learning and gathering of components. Also, in Week 2 and 3 of December starts the project's coding, design, and development, including construction and calibration. The beginning of January is the start of the debugging of the project, and the end of January is when the initial testing of the system starts. Along with this is the debugging of the project before its final testing. Finally, March will be the tentative date for the revision and release of the prototype.

Hardware Requirements

Raspberry Pi 5 (8GB)



Figure 5 *Raspberry Pi 5*

The Raspberry Pi 5 is the latest single-board computer from the Raspberry Pi Foundation, delivering a large CPU/GPU performance increase over the Pi 4 and improved I/O (PCIe, faster SD card performance, dual 4Kp60 micro-HDMI, USB 3.0). The 8GB model uses LPDDR4X memory and supports standard Raspberry Pi power and connectivity options.

Features	
	Detail
Processor	Broadcom custom SoC (Quad-core Arm Cortex-A76) @ 2.4 GHz.
Memory	8 GB LPDDR4X (model variant).
GPU / Video	VideoCore VII GPU; dual micro-HDMI outputs up to 4Kp60.
Storage	microSD slot (high-speed SD104 supported).
USB	2 × USB 3.0, 2 × USB 2.0.
Wireless	Dual-band 802.11ac Wi-Fi, Bluetooth 5.0 / BLE.
I/O	PCIe lane for add-ons, standard GPIO header, CSI/DSI connectors.
Power	USB-C power input (recommended 5.1V @ 5A for heavy loads).
Operating temperature	Standard SBC operating ranges (check product brief).

Table 4 *Raspberry Pi 5 (8GB) features*

24-inch 1080P Capacitive-Touch Computer Monitor



Figure 6 24-inch capacitive touch monitor

A 24-inch IPS touch monitor with capacitive multi-touch, 16:9 aspect ratio and 1920×1080 resolution. Ideal for kiosk or desktop use with HDMI/VGA/USB interfaces and VESA 100×100 mounting.

Features	
	Detail
Screen Size	24-inch
Panel Type	IPS
Touch Type	Capacitive Touch Screen
Screen Ratio	16:9
sRGB Color Gamut	100%
Interface Type	HDMI / VGA / USB
Resolution	1920 × 1080 (FHD)
Refresh Rate	75 Hz
Wall / VESA	100 × 100 mm
Viewing Angle	178°
Average Brightness	280 cd/m ²

Table 5 *24-inch 1080P Capacitive-Touch Computer Monitor Features*

Logitech S150 USB Stereo Speakers (model S150)



Figure 7 Logitech S150 USB Stereo Speakers (model S150)

Compact USB-powered stereo speakers for desktop use. Simple plug-and-play via USB-A for power/audio, with front panel volume/mute controls and an amber power indicator. Suitable for basic audio output in kiosk or desktop setups.

Features	
	Detail
Audio Power	Total watts (Peak) 2.4 W; Total watts (RMS) 1.2 W
Connection	USB-A (power and audio)
Controls	Volume up/down + mute on right speaker
Indicator	Amber LED on right speaker
Dimensions (each satellite)	H 158 mm × W 68 mm × D 64 mm (approx.)
Cables	USB power cable ~120 cm; speaker interconnect ~80 cm
Part number	980-000028 (Logitech S150 black).

Table 6 Logitech S150 USB Stereo Speakers (model S150) Features

Fifine K669B USB Condenser Microphone



Figure 8 *Fifine K669B USB Condenser Microphone*

USB condenser microphone with cardioid polar pattern designed for streaming, podcasting and voice capture. Plug-and-play USB connection, metal body with tripod stand and front volume/control knob.

Features	
	Detail
Type	Condenser, USB
Polar Pattern	Cardioid (uni-directional)
Frequency Response	20 Hz – 20 kHz
Sensitivity	~ -34 to -43 dB (manufacturer listing varies by SKU)
S/N Ratio	~78 dB
Power	5 V via USB
Connection	USB-A (plug-and-play)

Table 7 *Fifine K669B USB Condenser Microphone Features*

SECURE / 650VA UPS (Desktop UPS for PC)



Figure 9 *SECURE / 650VA UPS*

Line-interactive desktop UPS providing short backup runtime for small PCs and peripherals. Typical features include AVR (automatic voltage regulation), sealed lead-acid battery, simulated sine wave on battery, and ~10 ms transfer time. Use for ensuring clean shutdown of single-board computers and peripherals..

Features	
	Detail
Capacity	650 VA (typical VA → Watt conversion depends on unit; example: ~455 W rating in some listings)
Input Voltage	~220–230 VAC (region dependent)
Voltage Range	e.g., 140–300 VAC (model dependent)
AVR	Automatic Voltage Regulation (boost/cut ranges vary)
Output Waveform	Simulated sine wave (battery mode)
Transfer Time	Typical 4–10 ms (model dependent)
Battery Type	Sealed lead-acid, maintenance-free
Typical recharge	6–8 hours to ~90% after full discharge
Back-up time	Highly load dependent (e.g., ~10 min @ 75 W typical on some 650 VA units)

Table 8 *SECURE / 650VA UPS Features*

52Pi Case for Raspberry Pi 5 — Armor Lite V5 (with fan)



Figure 10 *52Pi Case for Raspberry Pi 5 — Armor Lite V5 (with fan)*

All-metal (CNC) aluminum heatsink case designed specifically for the Raspberry Pi 5. Integrates the Armor Lite V5 heatsink and a small fan interface (connects to the Pi’s JST fan header) to improve cooling and thermal headroom for heavy workloads. Allows access to SD, GPIO, camera/display connectors..

Features	
	Detail
Compatibility	Raspberry Pi 5 (designed to fit ports and connectors)
Cooling	Aluminum CNC heatsink + fan interface (fan included / optional)
Access	microSD, GPIO, camera/display connectors accessible
Material	Aluminum alloy (CNC machined)
Mounting / Fit	Designed for Pi 5 board dimensions; ventilation slots present

Table 9 *52Pi Case for Raspberry Pi 5 — Armor Lite V5 (with fan) Features*

52Pi Raspberry Pi 5 Power Supply — USB-C (5.1V @ 5A)



Figure 11 52Pi Raspberry Pi 5 Power Supply — USB-C (5.1V @ 5A)

Official/third-party 52Pi USB-C power adapter designed to supply the Raspberry Pi 5. Delivers 5.1 V @ 5 A and supports Power Delivery profiles for additional voltages (typical product listings show multiple outputs such as 9V/12V/15V at reduced currents). Includes built-in protections (OCP/OVP/SCP). Use the 5.1V@5A output for stable Pi 5 operation.

Features	
Feature	Detail
Input	100–240 V AC, 50/60 Hz
Output (main)	5.1 V @ 5 A (for Raspberry Pi 5)
Alternate PD outputs	e.g., 9 V @ 3 A; 12 V @ 2.25 A; 15 V @ 1.8 A (product dependent)
Protections	Over-current, Over-voltage, Short-circuit protections
Connector	USB-C
Power rating	~27 W max (typical for 5.1V@5A)

Table 10 52Pi Raspberry Pi 5 Power Supply — USB-C (5.1V @ 5A) Features

Software Requirements

Software requirements list all programs, libraries, frameworks, and tools needed to develop, deploy, and run the CoCo AI Kiosk. These ensure proper integration between the AI, frontend, backend, database, and hardware components.

Software	
Software / Tool	Purpose
Raspberry Pi OS	Operating system for Raspberry Pi 5
Python	Backend logic and API services
FastAPI	Framework for building RESTful APIs
React.js + TypeScript	Frontend web interface for kiosk
SQLite / PostgreSQL	Structured data storage for campus information
ChromaDB	Vector database for RAG knowledge base
OpenAI GPT API	AI language processing and response generation
Web Speech API	Speech-to-text and text-to-speech processing
Chromium Browser	Fullscreen kiosk display
Nginx	Web server for serving frontend and API requests

Table 11 *Software Requirements of CoCo*

The system's operating environment is Raspberry Pi OS, which provides support for both Python and frontend web technologies. The backend is developed using Python with FastAPI to handle API requests, process queries, and manage database operations. The frontend uses React.js with TypeScript for building the touchscreen interface and handling user interactions. The OpenAI GPT API provides natural language understanding and generation, while ChromaDB is used for indexing campus-specific knowledge in the RAG pipeline. The kiosk runs Chromium in kiosk mode for a full-screen user interface, and Nginx serves the web application and manages requests.

Database Requirements

The system's database serves as the central repository for campus-related information, enabling the AI module to deliver accurate and relevant responses. SQLite is used for lightweight storage of conversation history and session data, suitable for single-kiosk deployment. For more comprehensive and scalable operations, PostgreSQL stores structured campus information such as faculty names, office locations, consultation hours, building layouts, class schedules, event calendars, departmental directories, school policies, and student manuals.

The database also preserves institutional data, including vision, mission statements, core values, and school hymn. By maintaining a well-organized structure, the database ensures that the AI can efficiently retrieve, process, and present information to users. Conversation logs, including timestamps, user queries, and AI responses, are also stored for monitoring performance, analyzing user interactions, and improving system reliability.

3.2 Design

Through software that was used to design diagrams and 3-dimensional scales, the system applied the following diagrams to ensure the wellness of the system.

Block Diagram

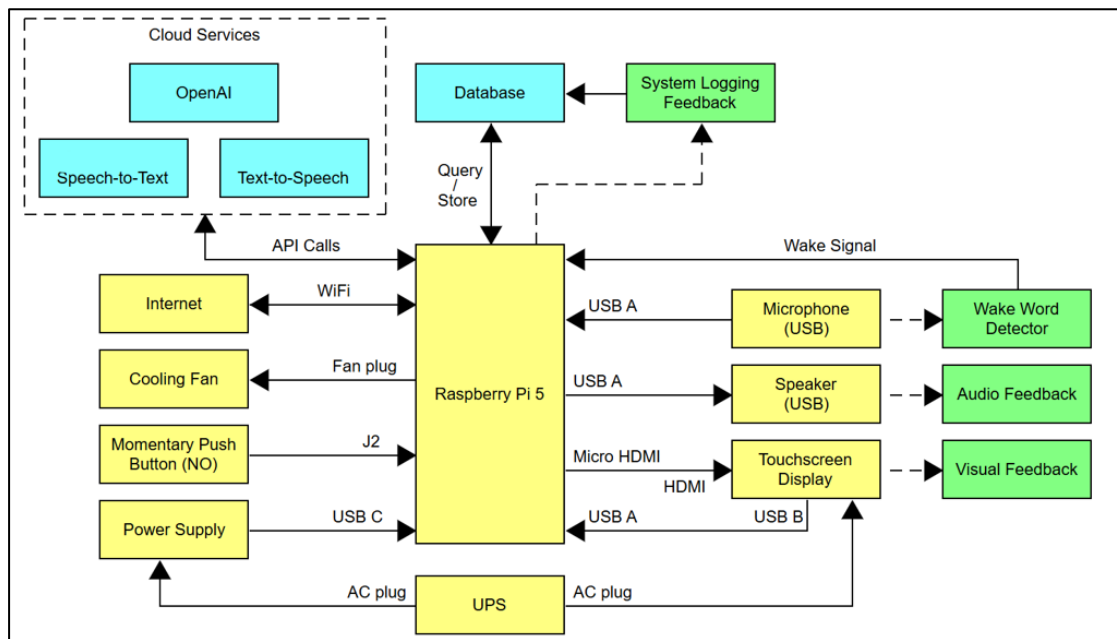


Figure 12 Block Diagram

Figure 12 illustrates the overall system architecture, showing the interaction between the Raspberry Pi 5, its peripheral devices, and external cloud services. The diagram outlines the flow of power, data, and control signals, including API communication with cloud-based speech processing services, database access, and system feedback. Connected input and output devices provide wake word detection, audio output, and visual feedback, while the UPS and power components support stable operation.

Use Case Diagram

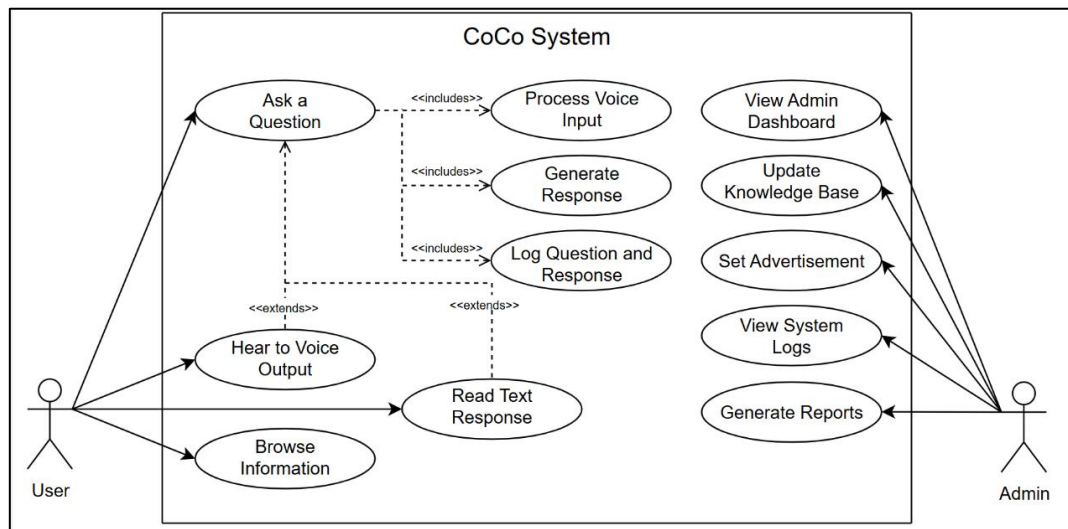


Figure 13 Use Case Diagram

Figure 13 shows the use case diagram of the CoCo System. The interaction begins when a user accesses the CoCo kiosk to initiate an inquiry through the “Ask a Question” function. The system then processes the user’s voice input, converts it into text, and forwards the information to CoCo’s response-generation module. After the system generates a suitable reply, it is either presented to the user through on-screen text or delivered using voice output. During this process, the system also logs the question and response for monitoring and future reference. Users may also browse general information available within the system interface.

On the administrative side, the admin accesses the system through the “View Admin Dashboard” use case. From here, the admin can update the knowledge base, configure or set advertisements, view system logs, and generate reports related to system performance and user interactions. These administrative use cases support system maintenance and ensure that CoCo continues to deliver accurate and up-to-date information to its users.

Activity Diagram

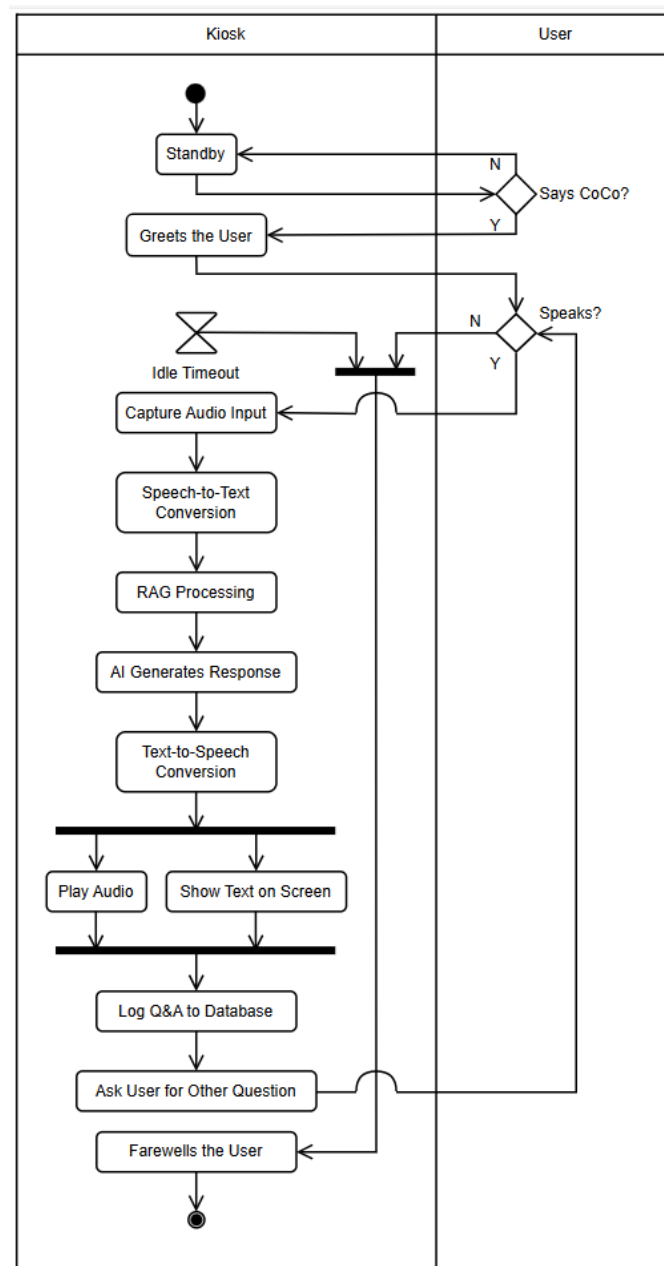


Figure 14 Activity Diagram

Figure 14 presents the activity diagram of the CoCo System. The process initiates with the kiosk in a standby state while waiting for the user to say the wake word “CoCo.” Once triggered, the system greets the user and transitions to an active listening phase. The kiosk captures the user’s voice input and converts it into text

through its speech-to-text component. The processed text is then forwarded to the RAG module and AI response generator to produce an appropriate answer. After generating the output, the system performs text-to-speech conversion and simultaneously displays the response on the screen and plays the audio through the speaker.

The system then logs the question-and-answer interaction into the database and prompts the user for any additional inquiries. If the user has no further questions, the kiosk delivers a farewell message and returns to standby mode, completing the activity flow.

Sequence Diagram

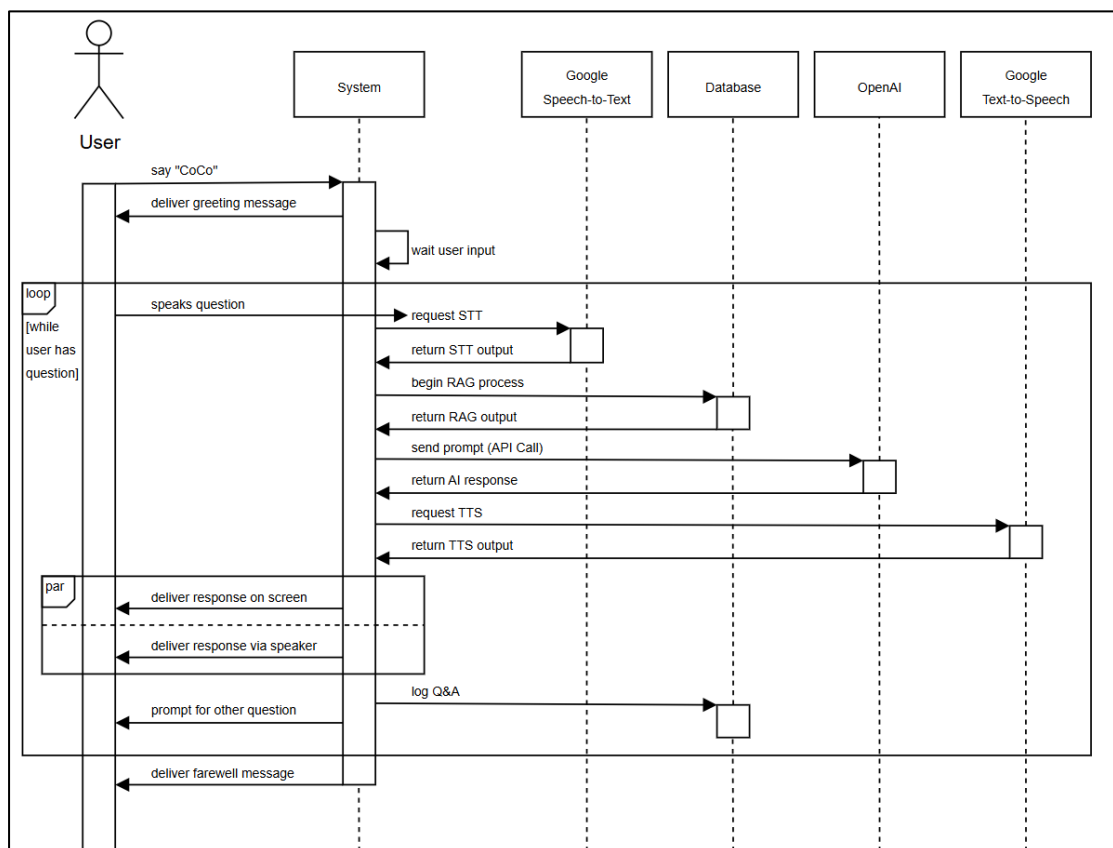


Figure 15 *Sequence Diagram*

Figure 15 illustrates the sequence diagram of the CoCo System. The interaction begins when the user says the wake word, prompting the system to deliver a greeting and wait for the user's input. Once the user speaks a question, the system sends the audio to the speech-to-text service, which returns the converted text. The system then initiates the RAG processing stage by retrieving relevant knowledge, after which the consolidated prompt is submitted to the AI model. Upon receiving the AI-generated response, the system forwards it to the text-to-speech service to obtain the audio output. The final response is delivered to the user both visually on the screen and auditorily through the speaker. The system logs the exchange into the database and then prompts the user for any additional questions. When no further interaction is detected, a farewell message is delivered, marking the end of the sequence.

Structural Design

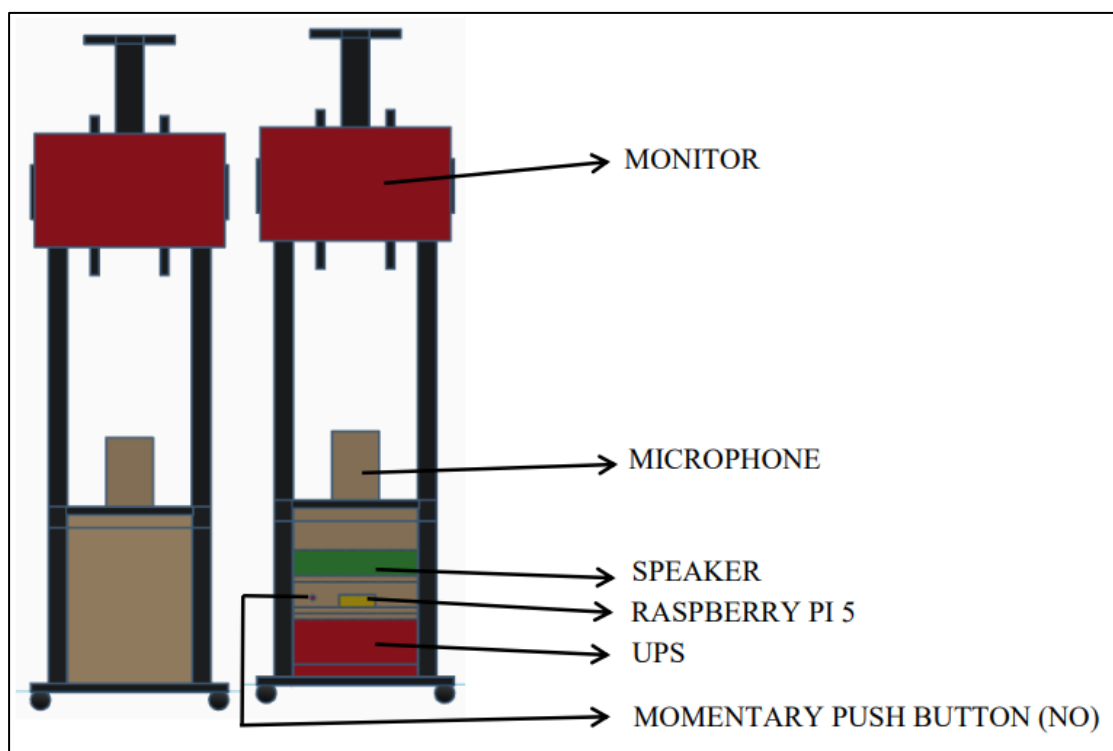


Figure 16 *Front View*

Using the online drawing software TinkerCAD, Figure 16 to Figure 19 presents the 3D structural model of the CoCo system. The frame or stand is designed with metal supports to ensure stability and accommodate all integrated components. The upper section houses the monitor, securely mounted to provide a clear user interface.

The middle section of the structure, which contains the microphone for voice input. The lower compartment encloses the speaker, Raspberry Pi 5, and an uninterruptible power supply (UPS), all positioned for efficient wiring and ventilation. A momentary normally open (NO) push button is mounted externally to allow system activation. The casing materials are arranged to protect electronic components while maintaining accessibility for maintenance.

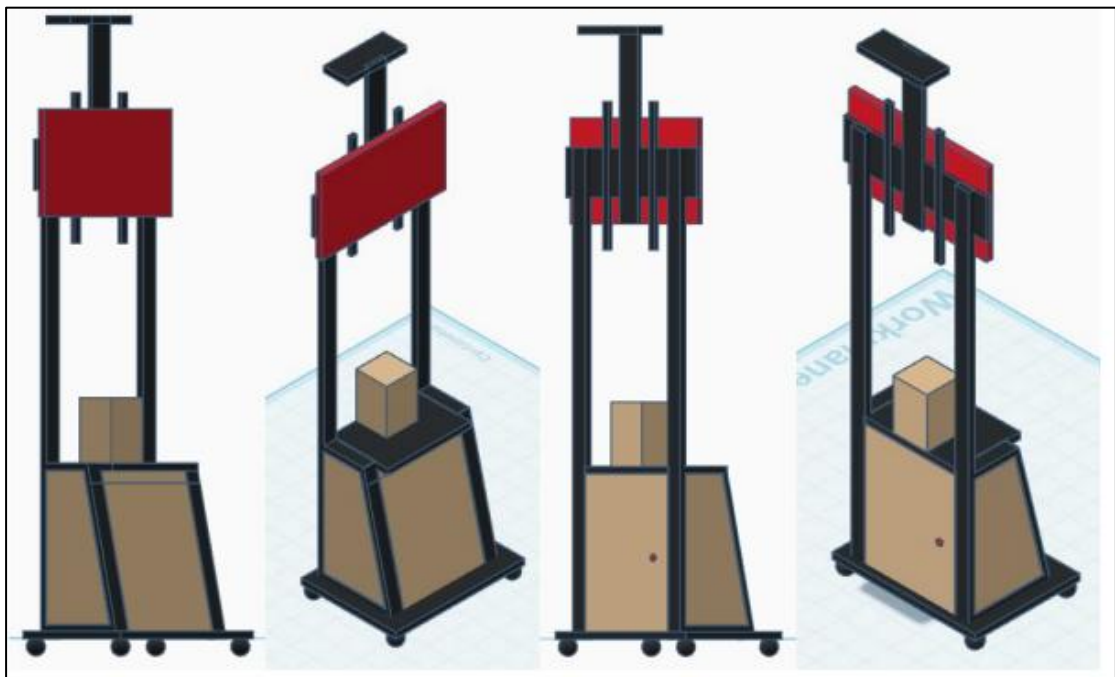


Figure 17 *Isometric View*

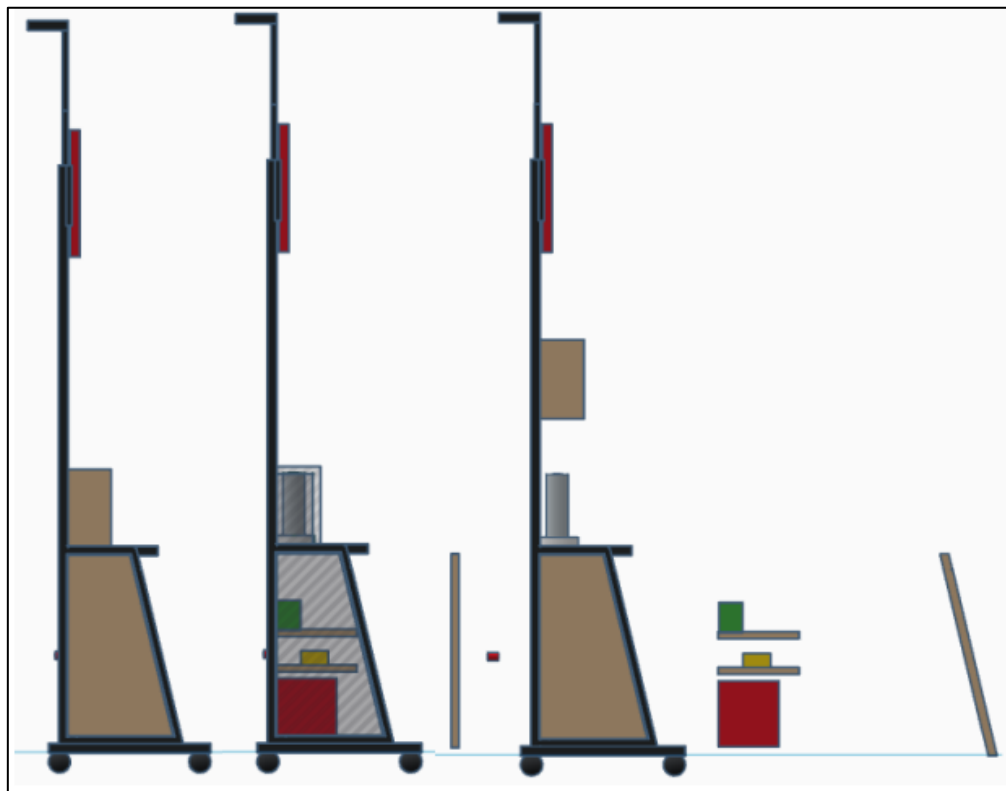


Figure 18 *Side View*

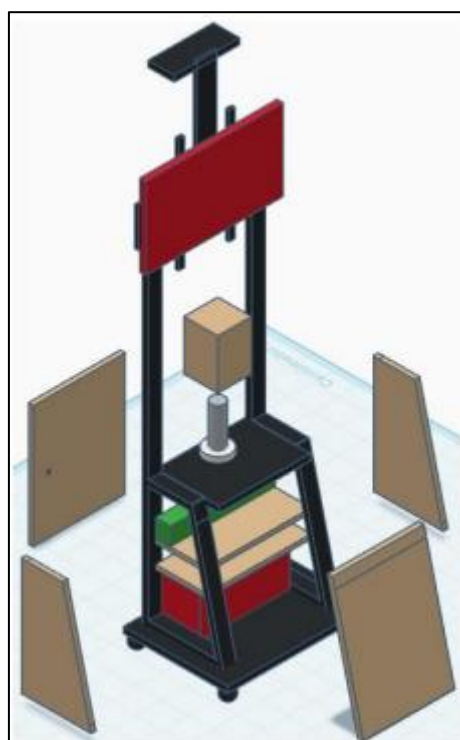


Figure 19 *Exploded View*

Schematic Diagram

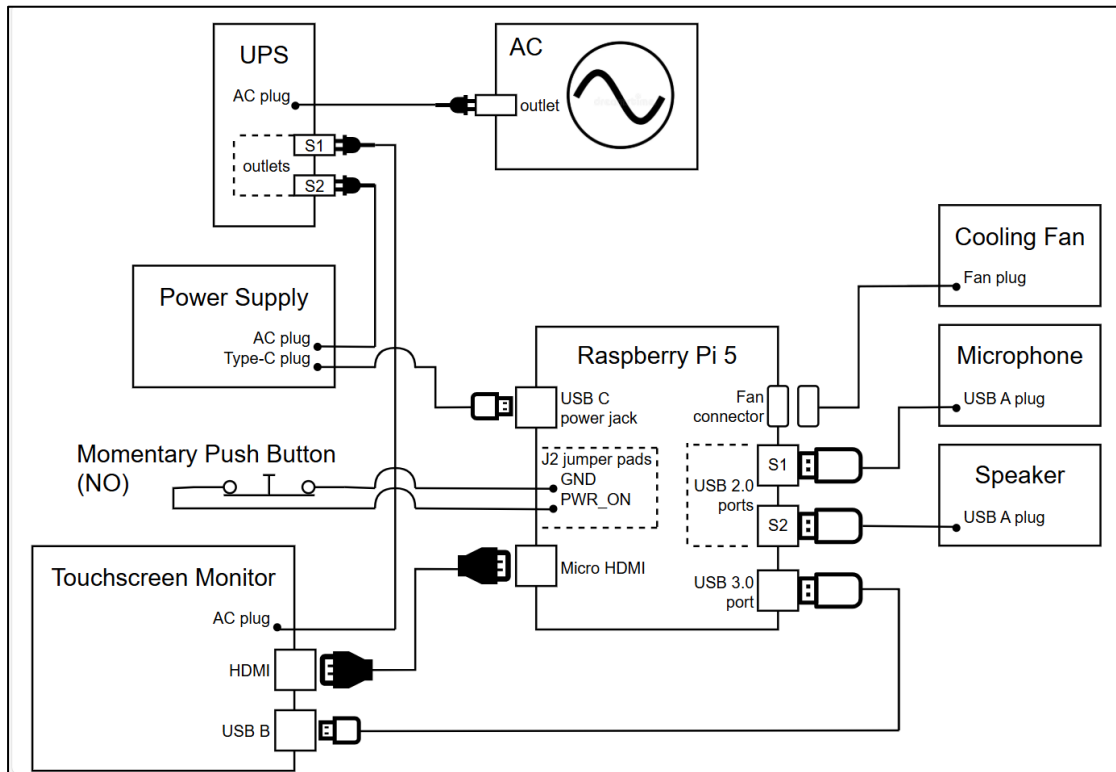


Figure 20 *Schematic Diagram*

Figure 20 depicts the system schematic diagram as created using an online diagram software called Draw.io. The diagram shows the system architecture, showing the interconnection of the Raspberry Pi 5 with its supporting peripherals and power components. The Raspberry Pi receives primary power through a USB-C supply routed via a UPS, while a momentary normally open push button interfaces with the J2 jumper pads to enable controlled power-on functionality. Peripheral devices, including the cooling fan, microphone, and speaker that are connected through the designated USB and fan headers. A touchscreen monitor interfaces through HDMI and USB, providing both display and touch input capabilities. All components are integrated to ensure stable operation under continuous AC or UPS-backed power conditions.

Component Diagram

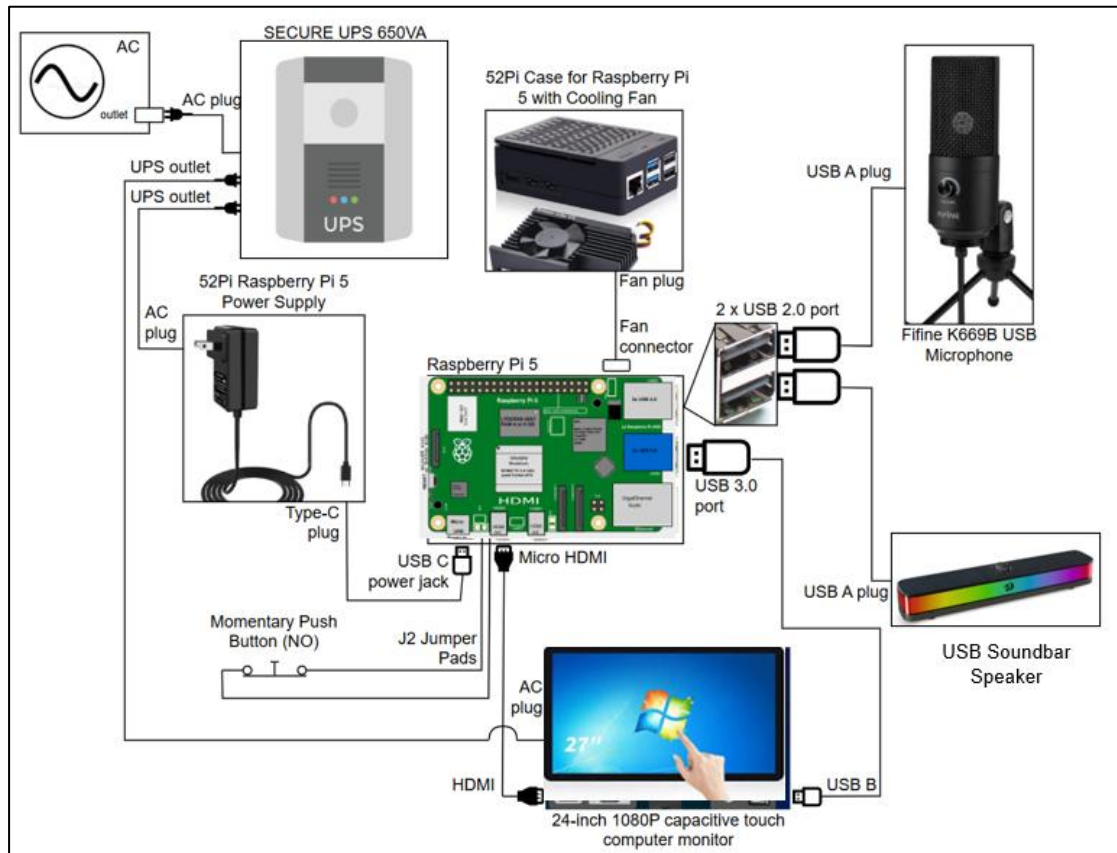


Figure 21 *Component Diagram*

Figure 21 presents component diagram for the system's hardware, detailing the Raspberry Pi 5 and its connected peripherals. The Raspberry Pi receives power through a Type-C supply linked to a UPS for continuous operation. A momentary normally open push button is connected to the J2 jumper pads to enable controlled system activation. The case fan is attached to the dedicated fan connector, while external devices such as the USB microphone, USB soundbar speaker, and touchscreen monitor interface through the Pi's USB and HDMI ports. All components are shown in relation to their power and data connections to provide a clear overview of the complete hardware configuration.

3.3 Development

The Development Phase focuses on constructing all components of the CoCo AI Kiosk based on the approved design. This includes the development of the intelligent language-processing module, the kiosk interface, the backend API server, the RAG-enabled knowledge base, and the integration of hardware elements such as the Raspberry Pi, microphone, speakers, and touchscreen display. The goal of this phase is to produce a complete and functional prototype capable of understanding user queries, retrieving institutional information, and delivering responses through both text and speech.

AI and Natural Language Processing Module

The AI module serves as the central intelligence of the system. CoCo utilizes the GPT-4o-mini-API, a cloud-based large language model, to interpret queries and generate coherent, context-aware responses. To support campus-specific questions, a Retrieval-Augmented Generation (RAG) system was implemented. This combines GPT's reasoning capabilities with a local knowledge base indexed through Sentence Transformers (all-MiniLM-L6-v2), which generate 384-dimensional embeddings for semantic search.

Document processing tools such as pdfplumber and python-docx were used to extract text from school documents. These were divided into overlapping chunks and stored in Chroma, the vector database responsible for fast semantic retrieval. Through this pipeline, the system can respond with accurate information about departments, courses, offices, processes, and campus policies.

The module also manages speech interaction. User speech is converted into text through the browser-based Web Speech API, while the system's response is converted back into audible output via the Speech Synthesis API. This enables a natural voice-assisted interaction, allowing users to communicate with CoCo hands-free.

Frontend – Website Application Interface

The user interface was developed using React.js with TypeScript. This framework was chosen for its reactivity, modularity, and ability to handle real-time updates. The interface runs on Chromium in kiosk mode, providing a full-screen, distraction-free experience. It displays AI responses, guides users through available options, and serves as the primary medium for visual interaction.

The frontend incorporates TailwindCSS for styling and uses Vite as the build tool to ensure fast compilation and performance on the Raspberry Pi. Communication between the frontend and backend is established through RESTful API calls, enabling CoCo to process user queries immediately. The UI was designed to be simple, responsive, and consistent with kiosk usability principles.

Backend

The backend was built using Python 3.11 and FastAPI, which provides a lightweight and high-performance framework suitable for deployment on the Raspberry Pi 5. It acts as the main controller of system logic, including routing user requests, generating AI responses, retrieving relevant campus information, and processing semantic search results from the RAG system.

The backend makes use of key Python libraries such as fastapi, uvicorn, httpx, sentence-transformers, chromadb, and sqlalchemy. It serves as the communication

layer between the frontend interface, the OpenAI API, and the database systems. Additionally, function calling from the GPT-4o-mini model helps the backend intelligently choose whether a query should be answered using school documents, external web search, or direct language-model reasoning. System-level tasks such as service startup, background processing, and error recovery are managed through systemd on Raspberry Pi OS to ensure reliable operation.

Database

CoCo uses a hybrid database structure. PostgreSQL stores structured campus information such as departments, offices, services, events, and directory details. This allows the kiosk to retrieve factual and static information efficiently. Meanwhile, SQLite is used for temporary data storage such as conversation history and session tracking, providing lightweight local storage suitable for single-kiosk deployment.

The knowledge retrieval system is powered by Chroma, which stores vector embeddings generated by Sentence Transformers. Queries are processed using cosine similarity search to determine the most relevant document segments before being passed to the language model. The combination of SQL databases and vector retrieval ensures the system can handle both structured information and unstructured documents.

External information can also be retrieved through the Brave Search API, allowing the kiosk to provide answers beyond the local knowledge base when appropriate.

System Program Flow

The overall program flow begins when the user interacts with the kiosk through either touch or voice. If voice input is used, the Web Speech API transcribes the speech into text. This input is then sent to the backend, where the system evaluates whether the query should be answered using the RAG knowledge base, direct GPT reasoning, or external web search.

Once processed, the final answer is delivered back to the user interface, displayed on the screen, and voiced aloud. Each user interaction is handled in a separate session to maintain privacy and prevent mixing of previous queries.

On the hardware side, the Raspberry Pi boots the entire system automatically. Through systemd services, the backend server is started first, followed by Nginx, which serves the frontend. Chromium then launches in kiosk mode, making the kiosk ready for user interaction. Error-handling routines also ensure that essential services restart automatically if unexpected failures occur.

3.4 Testing

The Testing Phase evaluates the performance, accuracy, reliability, and usability of the CoCo AI Kiosk before full deployment. This stage ensures that all system components such as AI processing, voice recognition, user interface, and hardware will function according to the specifications defined in the earlier phases.

The evaluation process began with preliminary functional testing, where individual features such as voice activation, speech-to-text, response generation, and touchscreen interaction were assessed. Once each component passed its initial verification, integration testing was performed to observe how the modules behaved when combined. This included checking whether user queries were successfully transmitted from the frontend to the backend, validating the accuracy of database retrievals, and examining the timing and consistency of AI-generated responses.

After integration, the system underwent full system testing in a controlled environment to simulate real usage. During this stage, the kiosk was tested with various campus-related queries to determine whether the system could provide accurate, context-appropriate answers. The behavior of the system in different environmental conditions, such as varying levels of background noise, inconsistent user speech patterns, and fluctuating network stability, was also observed.

3.5 Implementation

The Implementation Phase focuses on deploying the CoCo AI Kiosk into its intended environment and configuring all components to operate reliably for daily use. This phase includes setting up the hardware, installing necessary software, configuring system services, and preparing the kiosk for user interaction.

Implementation began with preparing the Raspberry Pi 5, where the Raspberry Pi OS was installed along with all required system packages. The backend server, database, and RAG indexing mechanism were set up and configured to start automatically using systemd services. This ensures that the server runs consistently, even after unexpected shutdowns or power interruptions.

The frontend or user interface was then deployed using Nginx, which served the React-based kiosk website in fullscreen mode. Chromium was configured to launch automatically on startup in kiosk mode, providing a smooth and uninterrupted interface that prevents users from accessing unintended system areas. Audio drivers, microphone settings, and speaker output were similarly calibrated to meet the system's voice input and output needs. Once the software environment was stable, the hardware components were assembled. The display was mounted, the USB microphone and speakers were connected, and the kiosk enclosure was arranged to maintain a clean and user-friendly appearance. Each component was tested to ensure proper functionality, including audio clarity, touchscreen responsiveness, and stable network connectivity.

The final step in implementation involved deployment within the campus environment. The kiosk was installed in a strategic location where students typically seek information. A trial run was conducted to observe real user interaction in a live

setting. Minor adjustments were performed based on early observations, such as refining the wake-word sensitivity, updating certain campus details in the database, and improving some UI text for clarity.

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